

CS 312 - Project Phase – 1 Report

Group Details:

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Phase-1 Details:

Project Title	Employee Database Management System (EDMS)
Phase -1 Components	• Set up GitHub Repo and Development Environment (Node, PostgreSQL). • Create PostgreSQL schema for employee records. • Implement user authentication routes (sign up / sign in). • Set up Express backend and test routes with Postman. • Develop React components: Login, Signup, Dashboard skeletons.

Task Status:

Task No	Task Details	Task Status
1	Initialize GitHub repository and configure local development environment • Create a new private GitHub repository named edms-project. • Clone it locally and set up project structure with /client (React) and /server (Node/Express). • Install required dependencies	Completed

2	Design and create PostgreSQL database schema for employee records - Define employees table with fields: id, first_name, last_name, email, position, role, created_at, updated_at. - Use pgAdmin to create the database. - Include a users table for authentication with hashed passwords and role flags.	Completed
3	Implement authentication API: Sign Up and Sign In endpoints - Create Express routes: POST /signup and POST /signin. - Use bcrypt to hash passwords before saving to DB and express-session to track logins. - On successful login, store user info and role in the session.	Completed
4	Develop React components: Login and Signup forms - Build controlled form components using useState and fetch/axios to POST to backend. - Handle success and error (display error messages).	Completed
5	Create base Dashboard component (Admin and Employee shell) - Implement a placeholder dashboard that conditionally renders content based on user role. - redirect if not authenticated.	Modified

New Tasks or Modified Tasks (Optional):

Task Type	Task Details	Task Status
Modified	Placeholder just on the signin and signup pages, dashboard redirects to signin. Pushed to phase 2 to make it more interactable.	Completed

Challenges and Solutions Implementation:

Details of the Challenge	Solution Implemented
Forgot to initialize the project via GitHub before starting local development	I created the full project locally first and later realized GitHub was not connected. I solved this by starting over and creating a new Repo.
Attempted signin using the hashed password instead of the original plain-text password	This caused repeated signin failures. I identified the mistake during manual bcrypt

	testing and put in the plain text password and was like “Duh”.
Postman tests initially failed because the backend was not running or misconfigured	Forgot to start the backend server before making requests. Once realized, launched the Express server using Node and verified routes were listening properly.
Missing express.json() caused req.body to be undefined during POST requests	Added the express.json() middleware before all route definitions in server.js to correctly parse JSON bodies, allowing proper access to req.body.

Phase-1 Developed Screens:

Task

Pictures

```
PS C:\Users\tyler\desktop\summer2025\cs312\edms\server> git remote -v
origin https://github.com/tjv55/Project_Group_tjv55_edms.git (fetch)
origin https://github.com/tjv55/Project_Group_tjv55_edms.git (push)
PS C:\Users\tyler\desktop\summer2025\cs312\edms\server> |
```

```
-- users table
CREATE TABLE users (
  id SERIAL PRIMARY KEY,
  email VARCHAR(255) UNIQUE NOT NULL,
  password VARCHAR(255) NOT NULL,
  role VARCHAR(20) NOT NULL CHECK (role IN ('admin', 'employee')),
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

-- employees table
CREATE TABLE employees (
  id SERIAL PRIMARY KEY,
  first_name VARCHAR(100),
  last_name VARCHAR(100),
  email VARCHAR(255) UNIQUE NOT NULL,
  position VARCHAR(100),
  role VARCHAR(20) NOT NULL,
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
  updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);
```

Sign In

Email		Password		Sign In
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Signup

test@admintest.co		••••••		Employee		Sign Up
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Signup successful!